

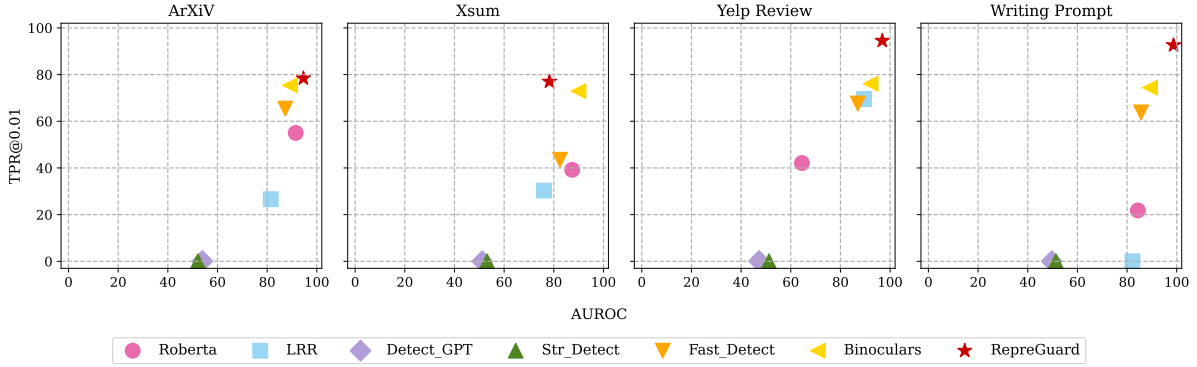


Figure 5: Performance Comparison of Various Detection Methods under OOD Domain Settings across Four Domain in Terms of AUROC (%) and TPR@0.01 (%) on a Test Set with 1000 "HWT-LGT" Pairs from 4 Different LLMs. The name of each subgraph corresponds to the test domain, while training is conducted on the other three domains. In each domain, the data consists of LGT from four LLMs.

5.1 Generalization on Domains

We evaluated our method on four domain datasets derived from different sources: ArXiv, XSum, Writing Prompt, and Yelp Review. As illustrated in Figure 5, most detectors exhibited poor performance in the OOD domain setting, especially with a low TPR@0.01. However, RepreGuard consistently achieves the highest average AUROC and TPR@0.01 on the OOD domain tasks, with 91.60% and 85.63%, respectively. Specifically, RepreGuard achieves the best performance in terms of AUROC and TPR@0.01 under the OOD domain settings for the Arxiv, Writing Prompt, and Yelp Review datasets. While its AUROC on the XSum dataset is slightly lower, RepreGuard still attains the highest TPR@0.01. In contrast, although the Roberta-based classifier, LRR, Fast-DetectGPT and Binoculars demonstrate strong performance in terms of AUROC, their TPR@0.01 values are quite low. For instance, the second-best detector, Binocular, achieved an average AUROC of 88.82% but a TPR@0.01 of 76.21%. This indicates that these detectors struggle to identify positive samples when operating at extremely low false positive rates, resulting in a higher risk of misdetections.

5.2 Detecting Texts with Varied Sizes

We evaluate the impact of text size on the performance of our detector. The results are shown in Figure 6. Overall, RepreGuard achieved the best performance on both short and long texts. Specifically, it attained an AUROC of 84.22% and a TPR@0.01 of 57.74% on short texts (64 tokens), while achieving an AUROC of 92.94% and a

TPR@0.01 of 81.70% on long texts (256 tokens). Although RepreGuard demonstrates slightly lower AUROC performance on 64-token texts compared to other detectors when trained on ChatGPT dataset, its TPR@0.01 consistently outperforms that of other detectors. Furthermore, as the text length increases, the performance advantage of RepreGuard becomes increasingly evident. On 256-token texts, its AUROC and TPR@0.01 significantly surpass those of other detectors, showcasing its exceptional capability in handling long texts. This indicates that RepreGuard remains effective in accurately identifying HWT, minimizing the risk of misclassifying it as LGT, even with shorter text sizes.

5.3 Robustness on Paraphrase & Perturbation Attack

We also evaluate the robustness of RepreGuard on mainstream attack methods, including paraphrase attacks and adversarial perturbation attacks. In practical applications, humans often make semantically equivalent revisions to LGT in line with their preferences. In addition, humans might intentionally introduce adversarial noise into LGT to evade detection, creating challenges for the detector. We used DIPPER Paraphraser (Krishna et al., 2023) and TextBugger (Li et al., 2019) to simulate these realistic scenarios, respectively. The results presented in Figure 7 and Figure 11 (see Appendix A.5) demonstrate that RepreGuard is the most robust detection method against both paraphrase and perturbation attacks. Significantly, this phenomenon is particularly evident under perturbation attacks, where the AUROC and TPR@0.01

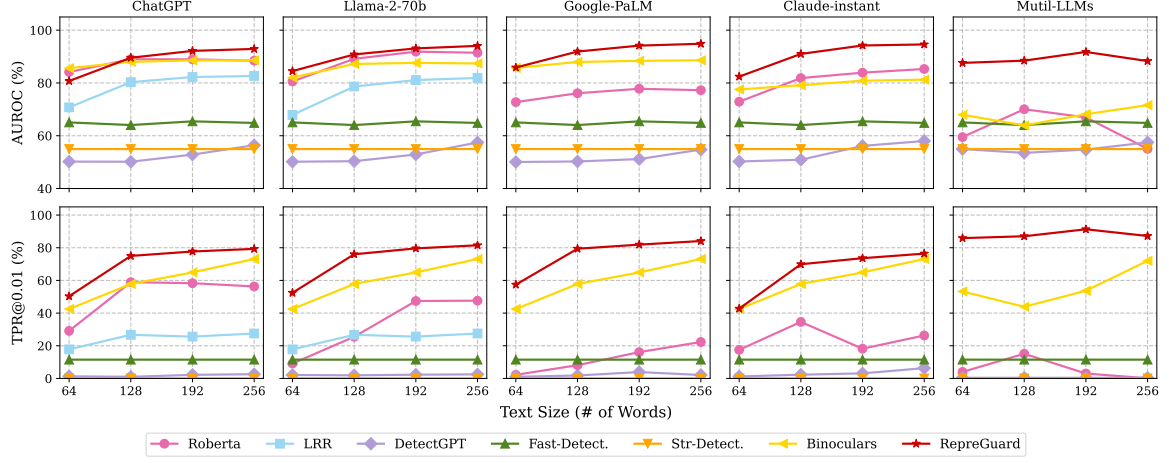


Figure 6: Performance Comparison of ReprGuard on Texts with Varied Sizes in Terms of AUROC (%) and TPR@0.01 (%) on a Test Set with 1000 “HWT-LGT” Pairs from 4 Different LLMs. The model name on each sub-graph refers to the LGT from different models used for representation features modeling and threshold setting.

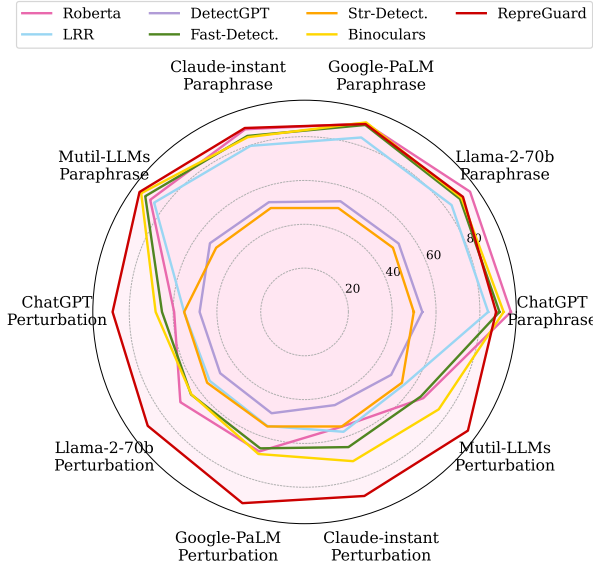


Figure 7: Performance Comparison of ReprGuard on Paraphrase and Perturbation Attack in Terms of AUROC on 1000 “HWT-LGT” Pairs from 4 Different LLMs. The raw text generated by each model is used to model representation features and set thresholds.

reach 89.65% and 88.63%, respectively, exceeding the second-best detector, Binocular, which achieves 69.45% and 58.54%. Additionally, although Roberta classifier performs well in certain aspects of AUROC, its TPR@0.01 is extremely poor, dropping as low as 0.10%, which highlights its significant limitations in identifying positive samples under strict thresholds. In contrast, while certain detectors exhibit strong performance on specific datasets and attacks, such as Fast-

DetectGPT achieving an AUROC of 89.70% on the Google-PaLM dataset under the perturbation attack, and LRR attaining an AUROC of 83.80% in paragraph attacks on the ChatGPT dataset, their overall performance remains poor, indicating that their robustness is significantly compromised.

5.4 Various Sampling Methods

Holtzman et al. pointed out that sampling strategies with maximum likelihood (such as beam search) often lead to text degeneration. Nucleus sampling addresses these issues by dynamically truncating the long tail of the probability distribution and sampling only from the “nucleus”. This approach effectively avoids degeneration, producing higher-quality and more diverse text, thereby making LGT closer to HWT. To investigate whether different sampling strategies would impact ReprGuard, we utilized the RAID dataset (Dugan et al., 2024) to evaluate the robustness of various sampling strategies. This dataset encompasses multiple domains and generative models and was constructed using diverse sampling approaches. Following the RAID setting, we also divided the data into Chat Models and Non-Chat Models and evaluated the AUROC metric. The results on Table 4 demonstrate that ReprGuard achieves the best performance across both models under the different sampling strategies, with an average of 6.42% in AUROC and 23.77% in TPR@0.01 higher than Binoculars. Noteworthy, most detectors, like LRR, Fast-Detect. and Binoculars perform well when the repetition penalty

Dec. Strategy	Chat Models (llama-chat, mistral-chat, mpt-chat)								Non-Chat Models (mistral, mpt, gpt2)								Avg.	
	greedy				sampling				greedy				sampling					
	Rep. Penalty?	X	✓		X	✓			X	✓			X	✓			AUR.	TPR.
Metrics	AUR.	TPR.	AUR.	TPR.	AUR.	TPR.	AUR.	TPR.	AUR.	TPR.	AUR.	TPR.	AUR.	TPR.	AUR.	TPR.	AUR.	TPR.
Roberta	88.32	65.97	83.58	41.02	88.57	46.81	71.66	13.27	93.11	40.22	74.45	14.87	80.24	<u>34.03</u>	<u>77.45</u>	<u>5.69</u>	82.17	32.74
LRR	90.17	49.00	80.23	12.69	86.13	24.25	67.17	4.19	94.86	87.43	83.43	34.33	77.25	0.40	50.00	0.20	78.66	26.56
Fast-Detect.	97.80	95.21	87.56	68.03	97.06	91.12	71.86	30.04	<u>98.65</u>	<u>96.21</u>	77.50	<u>48.10</u>	<u>84.03</u>	31.44	50.00	0.00	83.06	<u>57.52</u>
Str-Detect.	55.99	0.01	55.14	0.01	55.24	0.01	55.34	0.01	56.94	0.01	53.84	0.01	56.64	0.01	55.69	0.01	56.98	0.01
Binoculars	99.50	98.70	<u>91.52</u>	<u>71.26</u>	99.15	<u>94.41</u>	<u>77.69</u>	<u>31.24</u>	99.50	99.30	<u>79.54</u>	33.43	88.12	1.70	50.05	0.00	<u>85.63</u>	53.76
RepreGuard	<u>98.30</u>	<u>96.61</u>	97.16	94.81	<u>97.55</u>	94.61	94.86	85.73	98.50	92.22	92.47	75.55	72.55	34.43	81.99	46.31	92.05	77.53

Table 4: AUROC and TPR@0.01 for All Detectors Across Model Groups and Sampling Strategies. Sampling with a repetition penalty consistently makes most detectors difficult to detect, while RepreGuard maintains the best performance. The **Bold** indicates the best performance and underline indicates the second best.

mechanism is disabled. However, their AUROC showed a significant decline under the setting of the repetition penalty, whereas RepreGuard demonstrates strong robustness, with its performance only slightly decreasing and even improving on non-chat models. In contrast, most detectors experience a significant performance drop after enabling the penalty mechanism, especially in the sampling scenario of Non-Chat Models, where their detection capability almost completely deteriorates (AUROC approaching 50%). These indicate that RepreGuard can effectively detect the diversity and complexity of LGT by capturing internal representations, whereas LRR, Fast-Detect and Binoculars only capture information based on output probabilities, leading to the uncertainty introduced by different sampling strategies.

5.5 Costs of Space and Time

Detector ↓	AUR.	TPR.	Cost of Space	Cost of Time (Per sample)
Roberta	84.85	43.90	2.0GB	0.016s
Fast-Detect.	80.45	58.80	40.0GB	0.390s
Binocular	81.90	<u>72.60</u>	58.0GB	0.653s
RepreGuard (Phi-2)	96.10	54.50	<u>16.0GB</u>	<u>0.072s</u>
RepreGuard (Llama-3.1-8B)	<u>94.80</u>	77.10	38.0GB	0.359s

Table 5: Comparison of Effectiveness and Resource Costs on A Test Set with 1,000 ‘HWT-LGT’ Pairs from Four Different LLMs. These was trained on the Claude-Instant dataset with 512 ‘HWT-LGT’ pairs under the setting of NVIDIA A100 80GB using Float32 Precision. The **bold** indicates the best performance and underline indicates the second best.

When examining the costs of Methodology, we particularly focus on their balance between effectiveness and resource consumption in real-world applications. To evaluate our method in terms of effectiveness and resource cost, we compared RepreGuard with three other detectors: Roberta, Fast-Detect. and Binocular. In the comparative experiments, we set the batch size to 1 to measure the performance of each method when processing

a single sample in the setting of float32 under the A100 80GB GPU. The results in Table 5 shown that RepreGuard demonstrates the best overall performance with relatively low resource consumption. Specifically, RepreGuard (Phi-2) achieves the highest AUROC of 96.10% and relatively low resource consumption (16.0 GB, 0.072 seconds per sample), striking an effective balance between accuracy and efficiency. Meanwhile, RepreGuard (Llama-3.1-8B) achieves an AUROC of 94.80 and the highest TPR@0.01 of 77.10%, showcasing exceptional capability in positive case detection. It is noteworthy that Roberta lies in its extremely low resource consumption (2.0 GB, 0.016 seconds per sample), making it suitable for cost-constrained scenarios. However, its detection performance (84.85% in AUROC, 43.90% in TPR@0.01) is significantly inferior to that of RepreGuard.

In addition, we assess whether our approach is affected by memorization (see Appendix A.2), and examine how the performance as the increase of the LGT used in LLMs (see Appendix A.3)

6 Conclusion

In this paper, we introduce **RepreGuard**, a novel and reliable method based on hidden representation features for detecting text generated by LLMs. Experimental results on both ID and OOD, demonstrate RepreGuard’s strong detection capabilities and zero-shot proficiency. It requires only a small number of training samples to achieve impressive OOD generalization, effectively handling diverse real-world application scenarios and challenge from newly emerging LLMs. Furthermore, we verify the effectiveness, robustness, and generalization ability of RepreGuard in detecting texts of varied sizes, as well as texts that have undergone paraphrasing attack, perturbation attack, and various sampling methods.